

SHIZUHAMA AB, Japan

WMO: 476580

Lat: 34.813N

Lon: 138.298E

Elev: 10

StdP: 101.20

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	0.8	-12.1	1.3	4.7	-10.1	1.6	4.4	12.1	8.4	11.2	8.0	4.2	270	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	32.2	26.0	31.1	25.9	30.1	25.4	27.1	30.2	26.6	29.6	26.2	28.9	6.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.6	28.8	25.9	21.2	28.6	25.1	20.3	27.9	85.6	30.5	83.4	29.8	81.4	29.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	9.5	8.6	-2.6	35.0	1.0	1.5	-3.3	36.1	-3.9	37.0	-4.4	37.8	-5.1	38.9		
			-5.0	28.1	0.8	0.7	-5.6	28.6	-6.1	29.0	-6.5	29.4	-7.2	30.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	6.5	7.3	10.1	14.6	18.9	22.1	26.0	27.3	24.3	19.4	14.1	9.0
	DBStd	7.52	2.40	2.72	2.85	2.95	2.16	2.18	2.22	1.80	2.40	2.52	2.84	2.84
	HDD10.0	287	111	84	34	2	0	0	0	0	0	2	54	
	HDD18.3	1501	366	309	254	114	18	1	0	0	17	131	291	
	CDD10.0	2727	3	8	38	142	276	364	496	535	428	291	124	22
	CDD18.3	898	0	0	0	4	36	115	237	277	178	49	2	0
	CDH23.3	6677	0	0	0	14	110	430	1982	2772	1211	155	4	0
	CDH26.7	1813	0	0	0	1	12	74	561	893	256	17	0	0
Wind	WSAvg	3.7	4.6	4.3	4.2	3.9	3.4	3.2	3.3	3.2	3.0	3.1	3.4	4.4
Precipitation	PrecAvg	2177	81	109	180	199	213	252	246	179	271	224	144	78
	PrecMax	2950	187	234	312	379	311	460	563	678	588	899	328	184
	PrecMin	1510	1	28	67	48	107	92	25	14	81	23	13	1
	PrecStd	415	50	59	60	79	65	81	136	125	127	166	75	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	18.8	21.0	25.1	28.2	31.1	33.9	34.9	32.2	28.8	23.8	19.1
		MCWB	10.8	12.7	13.6	16.8	19.3	24.2	25.8	26.4	25.3	21.8	18.0	15.2
	2%	DB	14.0	15.9	18.2	22.8	25.9	28.9	32.0	32.9	30.2	26.2	21.8	16.9
		MCWB	9.3	11.2	12.2	16.1	18.7	23.2	25.9	26.3	24.9	20.7	16.5	12.2
	5%	DB	12.8	13.9	17.0	21.1	24.2	27.0	30.9	31.8	29.1	25.0	20.1	15.2
		MCWB	8.0	9.0	12.2	15.8	18.6	22.4	25.9	26.2	24.3	20.2	15.5	10.5
	10%	DB	11.1	12.2	15.6	19.8	23.0	25.8	29.8	30.2	28.0	23.9	19.0	14.0
		MCWB	6.3	7.7	11.3	15.1	18.3	22.1	25.5	25.8	24.0	19.4	14.7	9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.3	16.3	20.9	23.3	25.6	27.6	27.9	27.1	24.1	20.2	16.6
		MCDB	14.2	16.9	17.8	22.2	24.9	28.7	30.5	31.1	29.5	26.2	21.8	18.2
	2%	WB	10.3	12.2	15.0	18.4	21.2	24.6	26.8	27.2	26.2	22.5	17.9	13.4
		MCDB	12.6	15.1	16.8	20.7	23.3	27.2	30.1	30.5	28.6	25.1	20.0	15.6
	5%	WB	8.4	10.3	13.4	17.2	20.1	23.7	26.4	26.7	25.6	21.2	16.7	11.6
		MCDB	11.2	12.5	15.9	19.6	22.6	25.9	29.4	29.8	27.9	23.9	19.1	14.1
	10%	WB	7.1	8.6	11.8	16.0	19.2	22.7	25.9	26.4	24.9	20.1	15.5	10.3
		MCDB	10.3	11.2	14.7	18.4	21.9	25.0	28.7	29.3	27.2	22.8	18.1	13.0
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.6	8.7	8.1	7.4	5.8	5.7	6.1	6.1	7.0	7.9	8.3
		MCDBR	9.7	10.4	10.7	10.0	9.4	7.9	7.7	7.7	7.2	8.1	9.1	9.2
		MCWBR	6.6	7.3	7.3	6.3	5.3	3.9	3.4	3.4	3.8	4.7	5.9	6.3
	5% WB	MCDBR	8.5	8.6	8.9	7.9	7.6	6.5	6.6	6.5	6.2	6.8	7.5	7.8
		MCWBR	6.8	7.2	7.6	5.9	5.2	3.7	3.4	3.3	4.0	4.8	6.1	6.4
Clear-Sky Solar Irradiance	taub	0.373	0.423	0.483	0.533	0.547	0.567	0.556	0.544	0.497	0.472	0.413	0.373	
	taud	2.269	2.130	1.979	1.904	1.936	1.958	2.037	2.059	2.162	2.150	2.234	2.292	
	Ebn at Noon	811	805	786	768	762	745	749	751	767	746	758	785	
	Edh at Noon	106	135	171	193	189	185	170	163	141	132	109	98	
All-Sky Solar Radiation	RadAvg	2.76	3.36	4.23	4.85	5.23	4.61	5.12	5.25	4.12	3.24	2.66	2.45	
	RadStd	0.18	0.28	0.27	0.51	0.54	0.45	0.70	0.42	0.36	0.33	0.23	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page